

# 視窗應用程式開發與常用 元作介紹 (2)

李東霖 博士

國立臺灣海洋大學 電機工程學系

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# 常用元件介紹(2)

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- 通用元件

1. RichTextBox
2. CheckListBox
3. ProgressBar
4. DateTimePicker
5. ListView
6. MaskedTextBox
7. MonthCalendar
8. NotifyIcon
9. NumericUpDown
10. ToolTip
11. TreeView

- 容器

1. Panel
2. FlowLayoutPanel
3. SplitContainer
4. TableControl
5. TableLayoutPanel

- 功能表與工具列

1. ContextMenuStrip
2. MenuStrip
3. StatusStrip
4. ToolStrip
5. ToolStripContainer

# RichTextBox元件常用屬性

- 常用屬性
  - 配置
    - ColumnCount：直行數量
    - Columns：設定行大小
    - RowCount：橫列數量
    - Rows：設定列大小
    - Size：元件預設大小

方法:

`richTextBox1.AppendText();`//添加文字

`richTextBox1.ScrollToCaret();`//將捲軸拉至最後插入位置

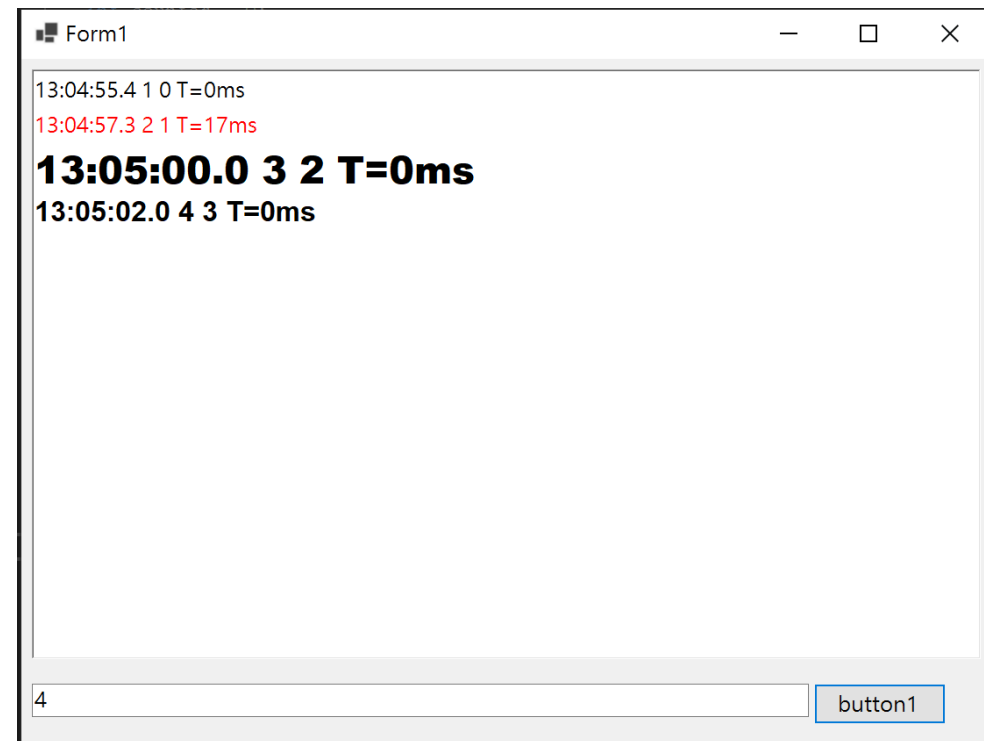
`richTextBox1.ResetText();`//將所有文字清除



# 時間戳記

using System.Diagnostics;

```
int counter = 0;
Stopwatch sw = new Stopwatch();
Font font1 = new Font("arial black", 16, FontStyle.Regular);
Font font2 = new Font("arial", 12, FontStyle.Bold);
private void button1_Click(object sender, EventArgs e)
{
    string last_elapseTime_ms = sw.ElapsedMilliseconds.ToString();
    sw.Restart();
    string timeStamp = DateTime.Now.ToString("HH:mm:ss.f ");
    string text = timeStamp + textBox1.Text + " " + counter
        + " T=" + last_elapseTime_ms + "ms\n";
    //string text_history = richTextBox1.Text;
    //text += text_history;
    //richTextBox1.ResetText();
    if (counter % 4 == 0) richTextBox1.SelectionColor = Color.Black;
    if (counter % 4 == 1) richTextBox1.SelectionColor = Color.Red;
    if (counter % 4 == 2) richTextBox1.SelectionFont = font1;
    if (counter % 4 == 3) richTextBox1.SelectionFont = font2;
    richTextBox1.AppendText(text);
    richTextBox1.ScrollToCaret();
    counter++;
    sw.Stop();
}
```



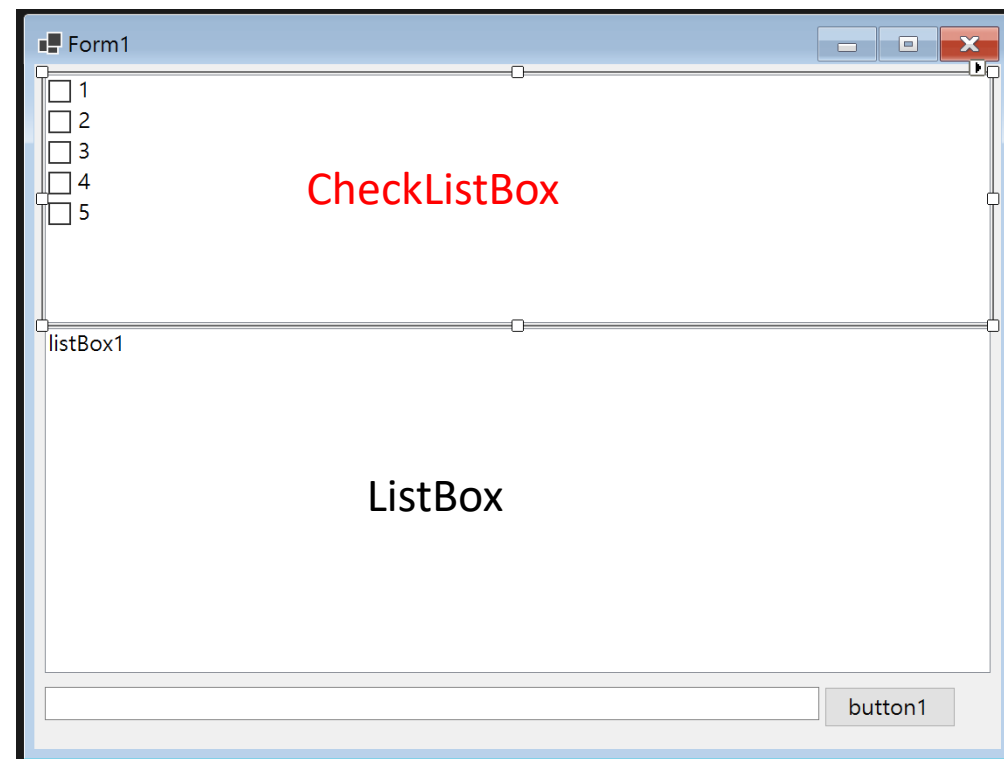
# CheckBox元件常用屬性

- 常用屬性

- 行為

- CheckOnClick：勾選或取消勾選時是否需連點
    - MultiColumn：是否一行多項目
    - ColumnWidth：一行多項目時寬度多少
    - Sorted：是否按字母排序

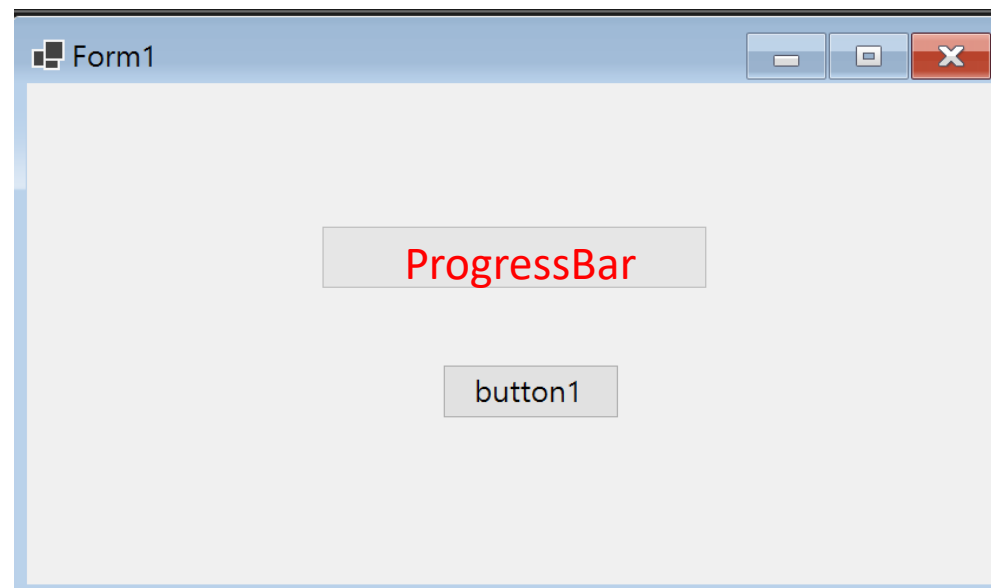
```
private void button1_Click(object sender, EventArgs e)
{
    checkedListBox1.Items.Add("new", true); //加入新項目並勾選
    for (int i = 0; i < checkedListBox1.Items.Count; i++)
        if (checkedListBox1.GetItemChecked(i))
            listBox1.Items.Add(checkedListBox1.Items[i].ToString());
}
```



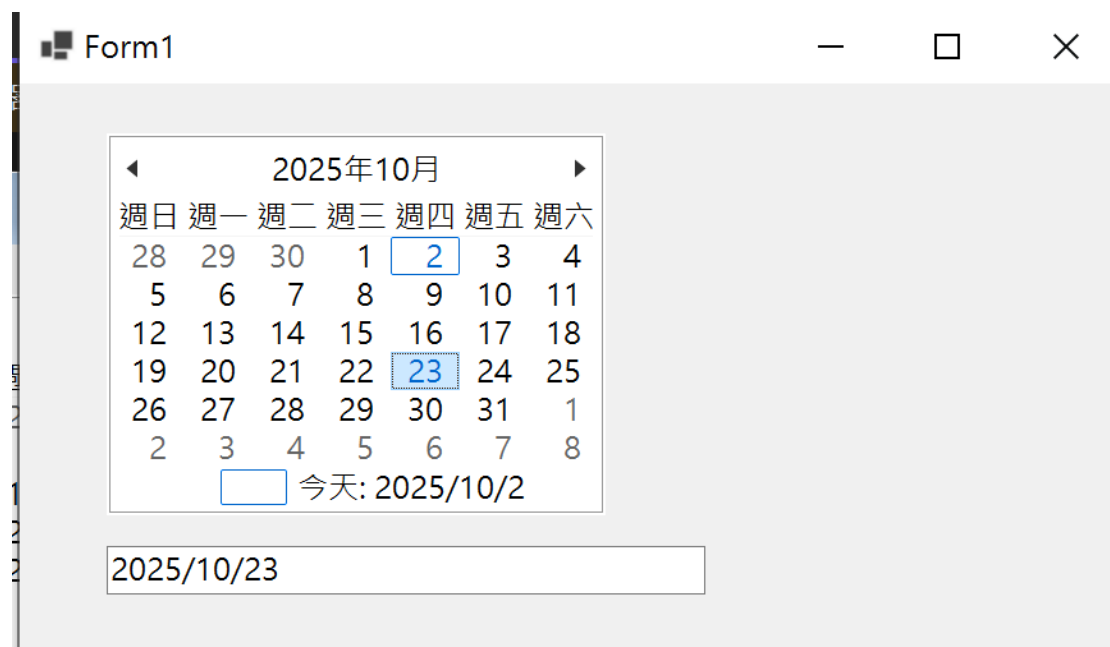
# ProgressBar元件常用屬性

- 常用屬性
  - 行為
    - Minimum：設定進度條的最小值。
    - Maximum：設定進度條的最大值。
    - Value：設定或取得目前的進度值。
    - Style：設定進度條的樣式，如連續、塊狀等。
    - Step：設定每次增加或減少的步長。

```
int counter = 0;
private void button1_Click(object sender, EventArgs e)
{
    progressBar1.Value = counter++;
    if (progressBar1.Value == progressBar1.Maximum)
    {
        MessageBox.Show("你總共按了" + progressBar1.Maximum.ToString() + "下");
    }
}
```



# MonthCalendar元件常用屬性

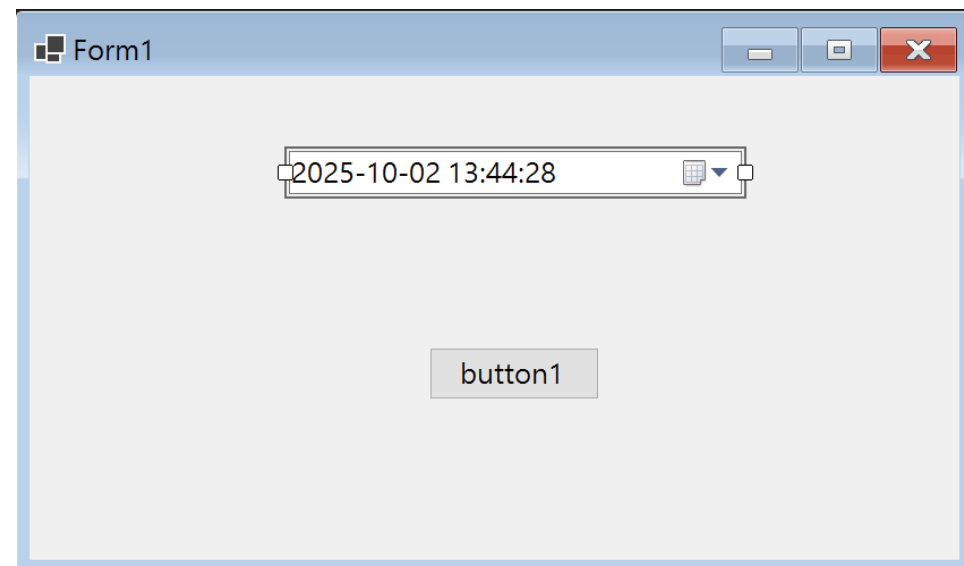


```
private void monthCalendar1_DateSelected(object sender, DateRangeEventArgs e)
{
    textBox1.Text = monthCalendar1.SelectionStart.ToShortDateString();
}
```

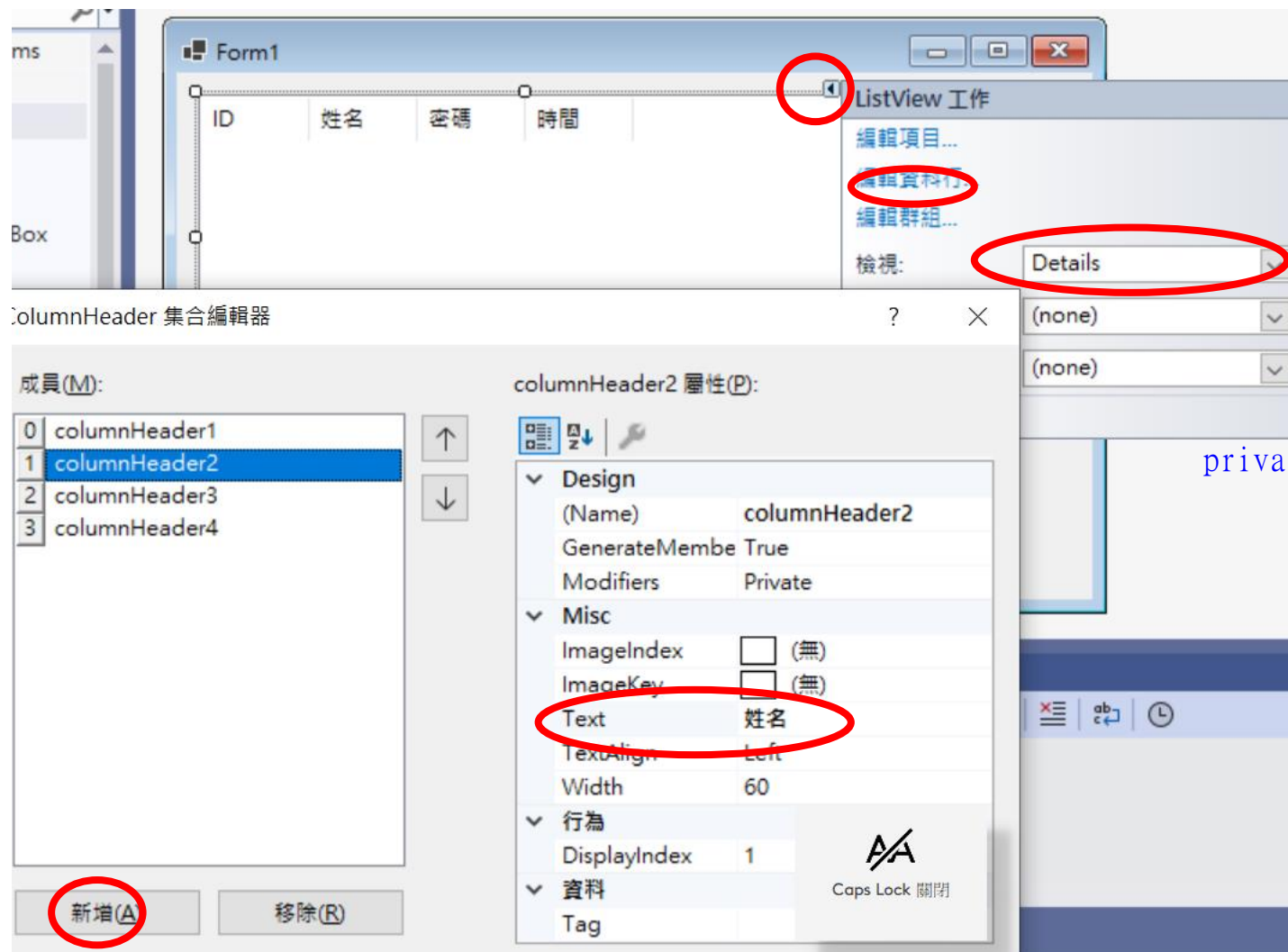


# DataTimePicker元件常用屬性

- 常用屬性
  - 外觀:
    - Format: 顯示格式
  - 行為:
    - CustomFormat: 自定顯示格式
- dateTimePicker預設只顯示日期
- 如要顯示時間
  - 將Format設為Custom
  - 將CustomFormat設為yyyy-MM-dd HH:mm:ss
  - 但時間部份只能透過鍵盤上下調整



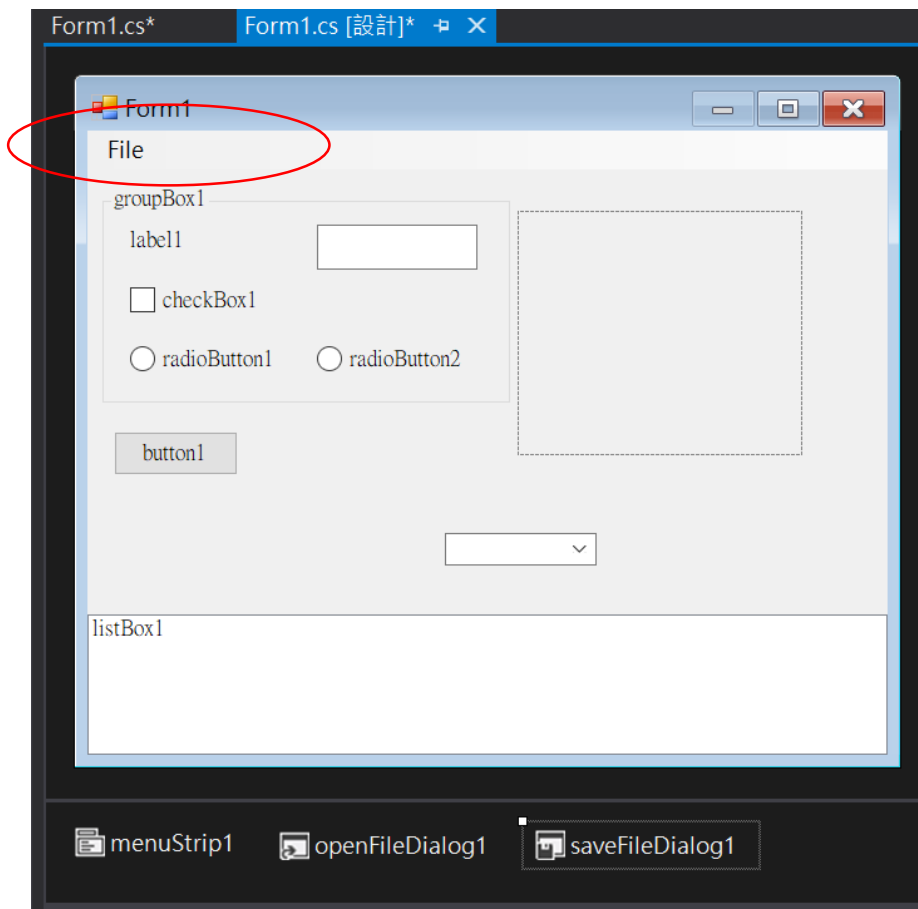
# Listview元件常用屬性



新增資料

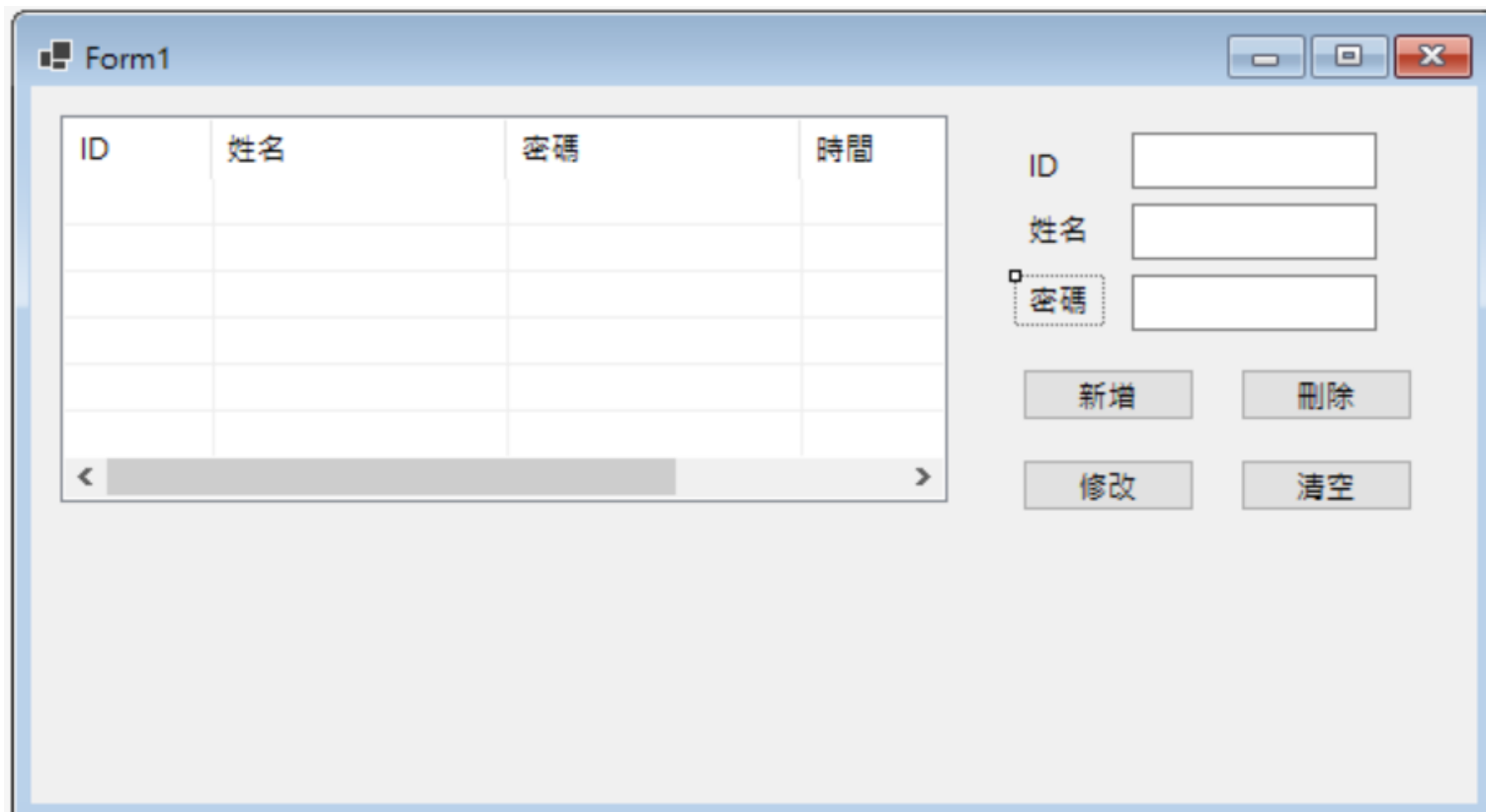
```
private void button1_Click(object sender, EventArgs e)
{
    string time = DateTime.Now.ToString();
    if (nameBox.Text != "" && pwBox.Text != "")
    {
        listView1.BeginUpdate();
        ListViewItem item = new ListViewItem();
        item = listView1.Items.Add(idBox.Text.Trim());
        item.SubItems.Add(nameBox.Text.ToString());
        item.SubItems.Add(pwBox.Text.ToString());
        item.SubItems.Add(time);
        listView1.EndUpdate();
    }
}
```

# MenuStrip元件常用屬性與應用



實習時間

# 作業 - 完成會員資料操作功能



The screenshot shows a Windows application window titled "Form1". Inside the window, there is a table with four columns: "ID", "姓名", "密碼", and "時間". The table has several empty rows. To the right of the table, there are three input fields labeled "ID", "姓名", and "密碼". Below these fields are four buttons: "新增" (Add), "刪除" (Delete), "修改" (Modify), and "清空" (Clear). The "刪除" button is highlighted with a dashed border.

| ID | 姓名 | 密碼 | 時間 |
|----|----|----|----|
|    |    |    |    |
|    |    |    |    |
|    |    |    |    |
|    |    |    |    |
|    |    |    |    |
|    |    |    |    |

刪除以姓名或ID為搜尋條件

修改以姓名為搜尋條件